MIL-H-46354A(MU)
21 October 1974
SUPERSEDING
MIL-H-46354(ORD)
20 September 1962

MILITARY SPECIFICATION

HEADRESTS, OPTICAL INSTRUMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - The specification covers headrests, browrests, and crash pads for use with Fire Control Instruments. These items collectively are referred to in this specification as headrests.

1.2 Classification.- Headrests shall be of the following types:

Type I - Encased mounting bracket
Type II - Adhered mounting bracket

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-S-735	Standard Test Fluids; Hydrocarbon

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and inspection of
MIL-P-14232	Parts Equipment and Tools for Army Material, Packaging and Packing of
MIL-F-45133	Fire Control Components and Assemblies
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance of

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STANDARDS

Federal

FED-STD-595	Colors, Requirements for
FED-STD-601	Rubber Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-289	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-293	Visual Inspection Guide for Cellular Rubber Items
MIL-STD-417	Rubber Composition Vulcanized General Purpose, Solid (Symbols and tests)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the processing activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. - Headrests furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time for opening of bids (see 4.2 and 6.2).

3.2 Materials. - Materials shall be as specified on the applicable drawing. Choice of elastomeric materials shall be at the option of the manufacturer except that natural rubber shall not be used for portions of the assembly exposed directly to the atmosphere.

3.3 Construction. - The headrests shall be either Type I or Type II, constructed to the form and dimensions specified on the drawing (see 6.1).

3.3.1 Type I. - Type I headrests shall consist of a cellular elastomeric pad molded through and around a perforated mounting bracket and coated with a minimum 0.02 inch thick elastomeric skin.

3.3.2 Type II. - Type II Headrests shall consist of a molded cellular elastomeric pad adhered to a steel plate and the entire assembly coated with a minimum 0.02 inch thick elastomeric skin.

3.3.3 Metal components. - Shall be in accordance with the. applicable drawings and Specification MIL-F-45133.

3.4 Flexibility. - The flexibility of the pad in terms of unit pressure required to produce a compression-deflection equal to 25 percent of the thickness of the pad, shall be 7 ± 3 pounds per square inch (psi) at room temperature, 70°F ± 8°F.

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3.5 Resilience. - The pad shall return to within 95% of its original shape within one minute following flexibility testing at room temperature.

3.6 Accelerated aging. - After exposure to a temperature of 158°F ± 2°F for a period of 70 hours, the headrest shall not become tacky or deformed. Upon return to room temperature, flexibility of the pad shall be 7 ± 4 psi for 25 percent compression deflection.

3.7 Low temperature flexibility.

3.7.1 Type I. - Type I specimens previously subjected to the accelerated aging test shall require a unit pressure of not more than 15 psi to attain 25 percent compression deflection at -40°F ± 2°F after thermal stabilization at this temperature for 20 to 24 hours.

3.7.2 Type II. - Type II specimens previously subjected to the accelerated aging test shall require a unit pressure of not more than 20 psi to attain 25 percent compression deflection at 40°F ± 2°F after thermal stabilization at this temperature for 20 to 24 hours.

3.8 Adhesive bond strength.

3.8.1 Coating to Pad. - The adhesive strength of the bond between the coating and the pad shall be greater than the cohesive strength of the pad or of the coating.

3.8.2 Pat to backplate. - Type II only. The adhesive strength of the bond between the pad and the backplate shall be greater than the cohesive strength of the pad.

3.9 Coating.

3.9.1 Tensile strength and elongation.- The coating material shall have a minimum tensile strength of 300 psi and a minimum ultimate elongation of 200 percent.

3.9.2 Fungus resistance. - The coating material shall be fungistatic or suitably treated to retard fungus growth.

3.9.3 Oil resistance. - The coating material shall comply with the oil resistance requirements~ percent volume change for Type S, Class S rubber in MIL-STD-417.

3.9.4 Color. - The color of the coating material shall be black comparable to Color Chip No. 27038 or 37038 of FED-STD-595.

3.9.5 Toxicity and reactions with human skin.- All portions of the fabricated headrest shall be inert and inherently incapable of causing a toxic or irritant effect as a result of contact with human skin.

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3.9.6 Workmanship. - Workmanship shall be of a quality consistent with the highest commercial standards. All material shall be sound, of uniform quality and condition, and free from structural defects. The coating shall be free of runs, blisters, pock marks and surface voids or inclusions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspction terms shall be as listed in MIL-STD-109.

4.2 Classification of inspection.- Inspection requirements specified herein are classified as follows:

a. Qualification inspection
b. Acceptance inspection

4.3 Qualificaton inspection. - A qualification sample of eight test specimens of each type specified shall be submitted for qualification testing in compliance with 3.1. Test specimens for Type I shall be in accordance with Figure I. Test specimens for Type II shall be in accordance with Figure II. The samples shall be representative of the materials and construction of the headrest proposed to be furnished under contract. Qualification testing shall be conducted by the Government at a place designated by the Government and shall consist of an examination for the characteristics specified in Table I in the order listed. The eight test specimens shall conform to the requirements specified for its type.

TABLE I - SEQUENCE OF QUALIFICATION TESTING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Flexibility	3.4	4.6.3
2.	Resilience	3.5	4.6.4
3.	Accelerated aging	3.6	4.6.5
4.	Low temperature flexibility		

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(a) Type I	3.7.1	4.6.6
(b) Type II	3.7.2	4.6.6
5. Adhesive bond strength		
(a) Coating to pad	3.8.1	4.6.7.1
(b) Pad to backplate (Type II only)	3.8.2	4.6.7.1
6. Coating		
(a) Tensile strength and elongation	3.9.1	4.6.8.1
(b) Fungus resistance	3.9.2	4.6.8.2
(c) Oil resistance	3.9.3	4.6.8.3
(d) Color	3.9.4	4.6.8.4
(e) Toxicity	3.9.5	4.6.8.5
(f) Workmanship	3.9.6	4.6.9

NOTE: Two specimens shall be selected for the fungus resistance test (see 4.6.8.2) from each of the eight specimens submitted. This shall be the only test required for these two specimens.

4.4 Acceptance inspection.

4.4.1 Submission of Product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.4.2 Classification of examination and tests.- Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table II herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section V herein. The tabulated classification of defects in Table II shall constitute the minimum inspection to be performed by the supplier after qualification approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level 11 of Table I with Sampling Plan Table II-A of Standard MIL-STD-105.

CRITICAL: None Defined
MAJOR: AQL 1.0 Percent Defective

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Construction	3.3	4.6.2
102. Flexibility	3.4	4.6.3
103. Resilence	3.5	4.6.4
104. Workmanship	3.9.6	4.6.9

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MINOR: AQL 4.0 Percent Defective

201.	**Color**	**3.9.4**	**4.6.8.4**
202.	**Identification marking**	**3.3, Dwg.**	**4.6.2**

NOTE: The above tests shall be conducted at 70°F ± 8°F temperature and in accordance with the test procedure paragraph listed.

4.4.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics, Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract-production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Materials	3.2	4.6.1
Accelerated aging	3.6	4.6.5
Low temperature flexibility	3.7	4.6.6
Fungus resistance	3.9.2	4.6.8.2
Toxicity	3.9.5	4.6.8.5

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

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4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of" the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Materials. - The headrest shall be visually examined for compliance with 3.2. The Government shall reserve the right to require a chemical analysis of materials or certification thereof, to insure quality of material in accordance with respective drawing or specification requirements specified herein.

4.6.2 Construction. - The headrest shall be subjected to a visual, tactile and dimensional inspection for conformance with the construction and configuration requirements of 3.3. Standard measuring equipment shall be used for dimensional inspections.

4.6.3 Flexibility.- The load applied to the specimen or headrest to determine compliance with 3.4 shall be made with a compression testing device. The device shall be equipped with a circular pressure foot with a surface contact area of 1.0 square inch. For contract areas that do not have this size of flat area, a pressure foot with an area of 0.5 square inch minimum shall be used. The device shall be capable of exerting a force of 7 ± 3 pounds per square inch (psi) on the specimen through a force gage with a minimum range of 10 psi, with an accuracy of 1/2 psi of full scale. The largest flat area of uniform thickness on the specimen shall be used as the test location. Apply the pressure gradually and in a perpendicular plane until the pressure foot has traveled 25 ± 2 percent of the free height of the specimen or headrest. The force required for this compressio shall be read from the force gage and shall not exceed the value specified. Force readings shall be taken 15 to 30 seconds after initial application of the force.

4.6.4 Resilience. - This test shall be conducted concurrent with the flexibility test. Upon removal of the applied load the specimen or headrest shall return to within 95 percent of its original shape within the time limit specified. Any deformation shall be measured with standard measuring equipment.

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4.6.5 Accelerated aging.- The aging test shall be performed in a heating chamber equipped with automatic temperature controls capable of maintaining the chamber at the operating temperatures specified and shall be suitably baffled to prevent dead spots and local overheating. The temperature variation throughout the chamber shall not exceed three degrees Fahrenheit. The heating medium shall be air, thoroughly circulated by mechanical means at atmospheric pressure. The source of heat shall be located in the air supply outside the chamber proper. Provisions shall be made for the vertical suspension of specimens without contact with each other or the sides of the chamber. A recording thermometer shall be located in the upper central portion of the chamber near the center of the specimens or headrests. The speciment of headrest shall be placed in the chamber, after the chamber has been pre-heated to a temperature of 158°F ± 2°F, and shall be kept in the chamber at that temperature for a period of 70 hours. At the end of the 70 hour period, the specimen or headrest shall be removed from the chamber and subjected to a tactile and visual examination for tackiness and deformation. The specimen or headrest shall then be cooled at room temperature (70°F ± 8°F) on a flat surface for a period of at least two hours. The specimen or headrest shall again be subjected to the flexibility test specified in 4.6.3. The force required for this compression deflection shall not exceed the value specified.

4.6.6 Low temperature flexibility (type I and type II). This test shall be performed in a cold chamber maintained at the specified low temperature. Subsequent to the oven aging test the specimen or headrest shall be placed in the cold chamber for a period of 20 to 24 hours. Immediately following this time cycle, and while the specimens are in the chamber, they shall be subjected to the flexibility test. The force required for a 25 percent compression deflection shall not exceed the value specified for the type of construction under test. The device and method of application used to perform the flexibility test of 4.6.3 shall be used for this test.

4.6.7 Adhesive bond strength.

4.6.7.1 Coating to pad.- The test specimen shall be a sample specimen of either Type I or Type II construction, as required, and shall be a specimen not previously used in other tests. This test shall be performed using a suitable tensile test device. Two parallel cuts one inch apart shall be made through the coating and one end of the strip thus formed shall be cut free and pulled up perpendicular to the specimen surface sufficient to form a gripping tab for the pull test. When pulled to failure at a rate of five inches per minute the rupture shall occur in either the pad or the coating and not in the bond (adhesive).

4.6.8 Coating.

4.6.8.1 Tensile strength and elongation.- This test shall be performed in accordance with Method 4111 of FED-STD-601. Sample test pieces shall be obtained by stripping the coating from the backplanes or pads of the test specimens used in the adhesive bond strength test of 4.6.7. The tensile strength and elongation of the coating shall meet the requirements of 3.9.1.

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4.6.8.2 Fungus resistance.- This test shall be conducted in accordance with MIL-F-13926 to determine compliance with 3.9.2. Subsequent to specimens meeting this test for qualification (see Table I) the Government reserves the right to require and conduct such independent tests on specimens as are considered necessary to determine continued compliance with 3.9.2.

4.6.8.3 Oil resistance.- This test shall be performed in accordance with Method 6211 of FED-STD-601. Sample test pieces shall be obtained by stripping the coating from the backplanes or pads of the test specimens used in the adhesive bond strength test of 4.6.7. The oil for immersion shall be Type VI of TT-S-735. The coating material shall meet the requirements specified in 3.9.3.

4.6.8.4 Color. - The test specimen or headrest shall be visually examined for correctness of color. Unless otherwise specified the color shall be a black comparable to Color Chips No. 27038 or No. 37038 of FED-STD-595.

4.6.8.5 Toxicity and reaction to human skin.- This test shall be a physical and visual test to determine compliance with 3.9.5. The headrest or specimen shall not cause a burning or itching sensation when held against the human skin at the forehead for a period of not less than three minutes in an ambient temperature between 93 to 95 degrees Fahrenheit. Also the material shall not become tacky after contact with the human skin for the duration specified above. In addition to the above, the contractor shall certify that the material used is not considered toxic, or known to cause dermatitis or other inflammatory reaction when in contact with human skin.

4.6.9 Workmanship. - The test specimens or headrests shall be visually inspected for conformance to the requirements of 3.9.6. The specimens or headrests shall be free from the specified defects. MIL-STD-289 and MIL-STD-293 shall be used as an inspection guide in this examination.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The headrest shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-P-14232 and the applicable packaging data sheet. The level of protection and data sheet shall be as specified in the procurement document (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type of headrest and applicable drawings.
c. Applicable stock number.
d. Applicable packaging data sheet and level of protection.

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6.2 Qualification. - With respect to products requiring qualification, awards will be made only for products which at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists is the U.S. Army Armament Command, Frankford Arsenal, Philadelphia, Pa. 19137, ATTN: SARFA-MDM. Information pertaining to qualification of products may be obtained from that activity.

Custodian:

ARMY-MU

Preparing Activity:

ARMY-MU

Project No. 1240-A622

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TEST SPECIMEN
TYPE I CONSTRUCTION

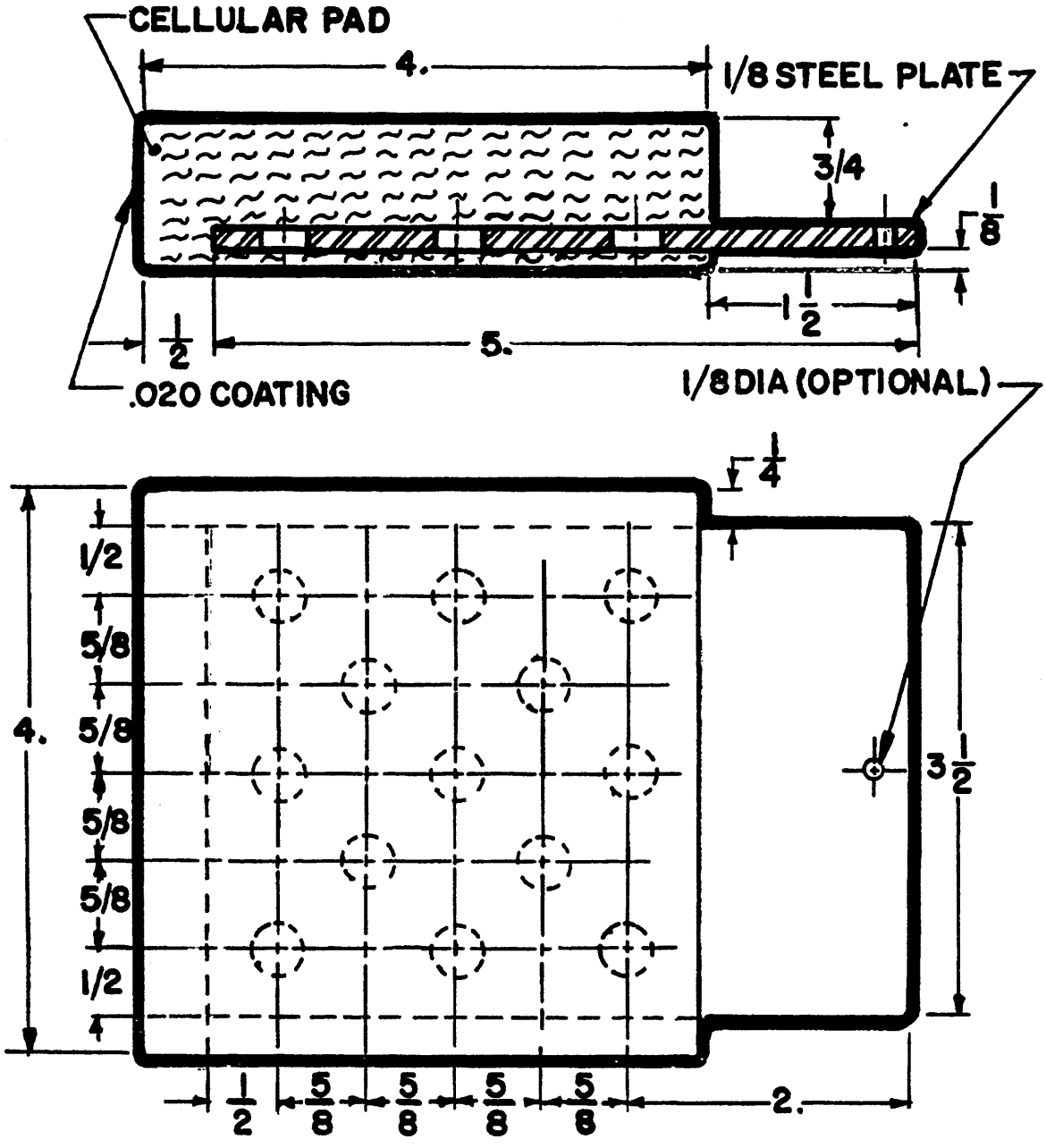


FIGURE I

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TEST SPECIMEN
TYPE II CONSTRUCTION

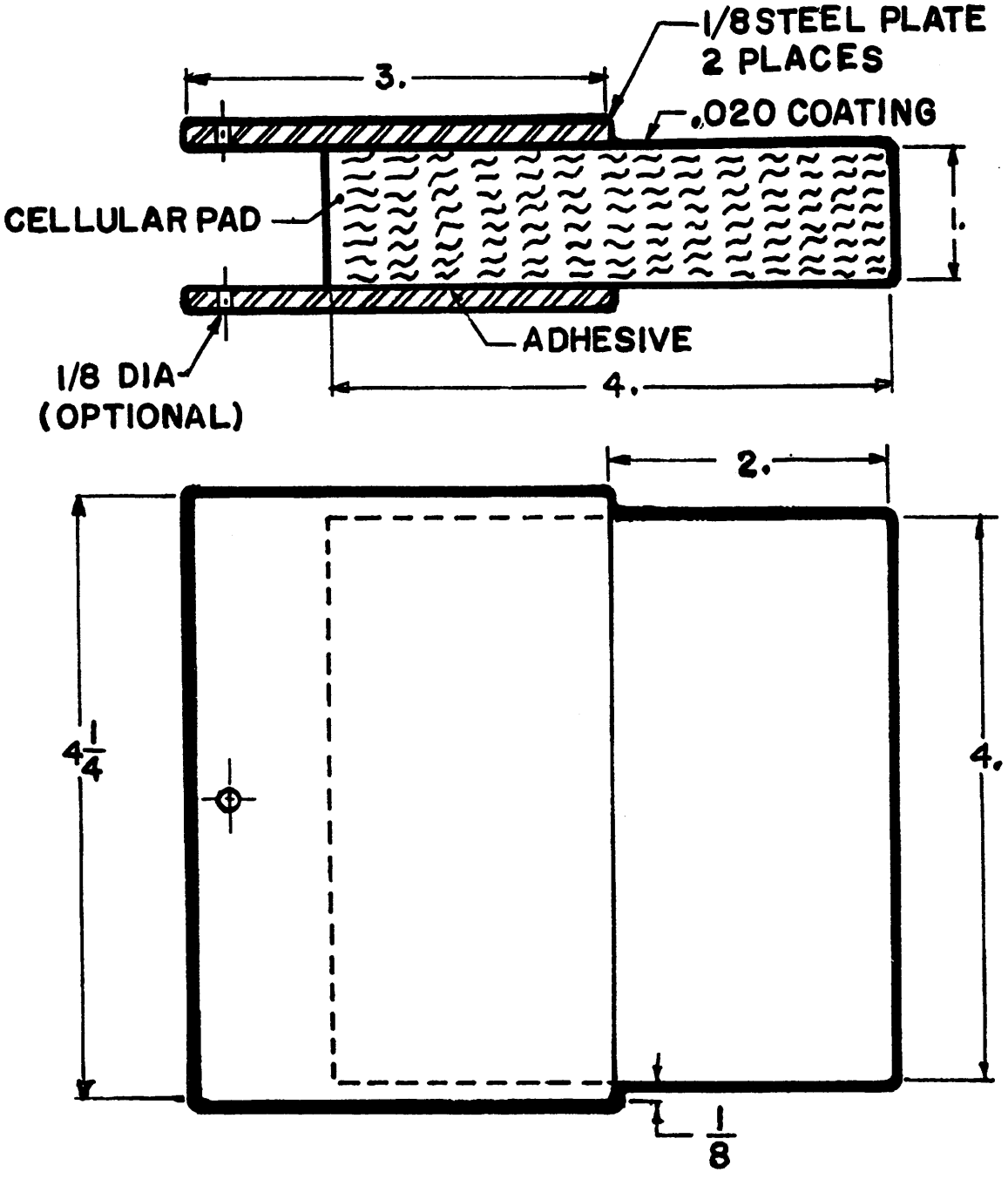


FIGURE II

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-H-46354A(MU) Headrests, Optical Instrument

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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